

Functions of Several Variables & Partial Derivatives



Evaluating functions of two variables

- 1 For $f(x, y) = x^2y - 3xy + y^2$, find $f(2, -1)$.
 - 2 For $f(x, y) = \sqrt{x^2 + y^2}$, find $f(3, 4)$ and describe the domain of f .
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Computing partial derivatives

- 3 For $f(x, y) = x^4y^3 + 2xy - y^5$, find f_x and f_y .
 - 4 For $f(x, y) = e^{x^2y}$, find f_x and f_y .
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Second-order and mixed partial derivatives

- 5 For $f(x, y) = x^3y^2$, find f_{xx} , f_{yy} , and f_{xy} .
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Clairaut's Theorem: equality of mixed partials

- 6 For $f(x, y) = \sin(xy) + x^2y$, verify that $f_{xy} = f_{yx}$.
 - 7 State Clairaut's Theorem and the condition it requires.
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Domains and level curves

- 8 Find the domain of $f(x, y) = \ln(x - y)$.
- 9 Describe the level curve $f(x, y) = 4$ for $f(x, y) = x^2 + y^2$.